Research Publications: Avinash Tiwari

Short author works

1. **A. Tiwari**, S. A. Bhat, S. J. Kapadia, T. R. Choudhury, S. Adhikari, and M. K. Singh, "Gravitational Wave informed inference of 21-cm global signal parameters," 2025, In preparation.
2. **A. Tiwari**, S. A. Bhat, M. A. Shaikh, and S. J. Kapaida, "Testing the nature of GW200105 by probing the frequency evolution of eccentricity," Sep. 2025, Accepted for publication in ApJ. DOI: `10.3847/1538-4357/ae1d74` arXiv: `2509.26152 [astro-ph.HE]`.
3. **A. Tiwari**, P. Chanda, S. J. Kapadia, S. Adhikari, A. Vijaykumar, and B. Dasgupta, "Profiling Dark Matter Spikes with Gravitational Waves from Accelerated Binaries," Aug. 2025, Under review in PRL. arXiv: `2508.03803 [hep-ph]`.
4. **A. Tiwari**, A. Vijaykumar, S. J. Kapadia, and S. Chatterjee, "Identifying and characterizing extragalactic circum-CBC exoplanets with future gravitational-wave detectors," 2025, In preparation.
5. **A. Tiwari**, A. Vijaykumar, S. J. Kapadia, S. Chatterjee, and G. Fragione, "Profiling stellar environments of gravitational wave sources," *Physical Review D*, vol. 112, no. 8, Oct. 2025, ISSN: 2470-0029. DOI: `10.1103/gspl-m478`
6. **A. Tiwari**, A. Vijaykumar, S. J. Kapadia, S. Ghosh, and A. B. Nielsen, "A pipeline to search for signatures of line-of-sight acceleration in gravitational wave signals produced by compact binary coalescences," Jun. 2025, Under review in PRD. arXiv: `2506.22272 [astro-ph.HE]`.
7. S. A. Bhat, **A. Tiwari**, M. A. Shaikh, and S. J. Kapadia, "EECT: an Eccentricity Evolution Consistency Test to distinguish eccentric gravitational-wave signals from eccentricity mimickers," Aug. 2025, Accepted for publication in PRD. DOI: `10.1103/rstg-6n6y` arXiv: `2508.14850 [gr-qc]`.
8. **A. Tiwari**, A. Vijaykumar, S. J. Kapadia, G. Fragione, and S. Chatterjee, "Accelerated binary black holes in globular clusters: forecasts and detectability in the era of space-based gravitational-wave detectors," *Mon. Not. Roy. Astron. Soc.*, vol. 527, no. 3, pp. 8586–8597, 2023. DOI: `10.1093/mnras/stad3749` arXiv: `2307.00930 [astro-ph.HE]`.
9. A. Vijaykumar, **A. Tiwari**, S. J. Kapadia, K. G. Arun, and P. Ajith, "Waltzing Binaries: Probing the Line-of-sight Acceleration of Merging Compact Objects with Gravitational Waves," *Astrophys. J.*, vol. 954, no. 1, p. 105, 2023. DOI: `10.3847/1538-4357/acd77d` arXiv: `2302.09651 [astro-ph.HE]`.

Collaboration Papers

1. A. G. Abac et al., "All-sky search for long-duration gravitational-wave transients in the first part of the fourth LIGO-Virgo-KAGRA Observing run," Jul. 2025. arXiv: `2507.12282 [gr-qc]`.
2. A. G. Abac et al., "All-sky search for short gravitational-wave bursts in the first part of the fourth LIGO-Virgo-KAGRA observing run," Jul. 2025. arXiv: `2507.12374 [astro-ph.HE]`.
3. A. G. Abac et al., "Directed searches for gravitational waves from ultralight vector boson clouds around merger remnant and galactic black holes during the first part of the fourth LIGO-Virgo-KAGRA observing run," Sep. 2025. arXiv: `2509.07352 [gr-qc]`.
4. A. G. Abac et al., "GW231123: a Binary Black Hole Merger with Total Mass 190-265 $M_\odot$," Jul. 2025. arXiv: `2507.08219 [astro-ph.HE]`.
5. A. G. Abac et al., "GW250114: Testing Hawking's Area Law and the Kerr Nature of Black Holes," *Phys. Rev. Lett.*, vol. 135, no. 11, p. 111 403, 2025. DOI: `10.1103/kw5g-d732` arXiv: `2509.08054 [gr-qc]`.
6. A. G. Abac et al., "GWTC-4.0: An Introduction to Version 4.0 of the Gravitational-Wave Transient Catalog," *arXiv e-prints*, arXiv:2508.18080, Aug. 2025, Art. no. arXiv:2508.18080. DOI: `10.48550/arXiv.2508.18080` arXiv: `2508.18080 [gr-qc]`.

7 A. G. Abac et al., "GWTC-4.0: Methods for Identifying and Characterizing Gravitational-wave Transients," Aug. 2025. arXiv: 2508.18081 [gr-qc].

8 A. G. Abac et al., "GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run," Aug. 2025. arXiv: 2508.18082 [gr-qc].

9 A. G. Abac et al., "Open Data from LIGO, Virgo, and KAGRA through the First Part of the Fourth Observing Run," Aug. 2025. arXiv: 2508.18079 [gr-qc].

10 A. G. Abac et al., "Search for Continuous Gravitational Waves from Known Pulsars in the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run," *Astrophys. J.*, vol. 983, no. 2, p. 99, 2025. DOI: 10.3847/1538-4357/adb3a0 arXiv: 2501.01495 [astro-ph.HE].

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13 "Black Hole Spectroscopy and Tests of General Relativity with GW250114," Sep. 2025. arXiv: 2509.08099 [gr-qc].

14 T. L. S. Collaboration, the Virgo Collaboration, and the KAGRA Collaboration, "Gwtc-4.0: Constraints on the cosmic expansion rate and modified gravitational-wave propagation," 2025. arXiv: 2509.04348 [astro-ph.CO]. URL: https://arxiv.org/abs/2509.04348

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16 "GWTC-4.0: Population Properties of Merging Compact Binaries," Aug. 2025. arXiv: 2508.18083 [astro-ph.HE].

17 G. Raman et al., "Swift-BAT GUANO Follow-up of Gravitational-wave Triggers in the Third LIGO–Virgo–KAGRA Observing Run," *Astrophys. J.*, vol. 980, no. 2, p. 207, 2025. DOI: 10.3847/1538-4357/ad9749 arXiv: 2407.12867 [astro-ph.HE].

18 A. G. Abac et al., "A Search Using GEO600 for Gravitational Waves Coincident with Fast Radio Bursts from SGR 1935+2154," *Astrophys. J.*, vol. 977, no. 2, p. 255, 2024. DOI: 10.3847/1538-4357/ad8de0 arXiv: 2410.09151 [astro-ph.HE].

19 A. G. Abac et al., "Observation of Gravitational Waves from the Coalescence of a 2.5–4.5 $M_\odot$ Compact Object and a Neutron Star," *Astrophys. J. Lett.*, vol. 970, no. 2, p. L34, 2024. DOI: 10.3847/2041-8213/ad5beb arXiv: 2404.04248 [astro-ph.HE].

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